

Caden Lippie

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EDUCATION

Syracuse University, School of Information Studies — Syracuse, NY

Aug 2025 – May 2026

Master of Science, Applied Data Science, AI Concentration — GPA: 3.88/4.0

Relevant Coursework: Deep Learning, Applied Machine Learning, Cloud Management, Advanced Big Data Management

Syracuse University, David B. Falk College of Sport — Syracuse, NY

Aug 2022 – May 2025

Bachelor of Science, Sport Analytics (3+1 Accelerated Program)

PROFESSIONAL EXPERIENCE

Data Science Graduate Assistant — Syracuse Women's Lacrosse, Syracuse, NY

Sep 2025 – May 2026

- Founded the program's first analytics department, defining its objectives and recruiting, interviewing, and supervising a cohort of 5 undergraduate data science interns.
- Directed intern projects in Python, including a pose-estimation tool measuring athlete running efficiency and an extension of the YOLOv8 tracking system (see Projects) to extract player positioning and formations from game footage.
- Led weekly project meetings and translated coaching staff requests into scoped analytics deliverables.

Data Science Intern — Syracuse Women's Lacrosse, Syracuse, NY

Jan 2023 – Aug 2025

- Designed and built a full-stack analytics platform (JavaScript/React, Python/FastAPI, SQL/PostgreSQL) serving real-time performance, wellness, and testing data to coaching staff for player evaluation and injury mitigation.
- Developed a data application in R (Shiny) with custom performance metrics; presented to coaching staff and iterated on feedback to drive adoption.

Data Science Intern — Kitman Labs (Remote)

Jan 2023 – May 2023

- Analyzed multi-factor datasets (travel, injury, box-score) in R to quantify the effect of scheduling on team performance for a leading sports data analytics platform.
- Presented findings to senior staff and co-authored a research paper documenting results (under NDA).

PROJECTS

Cognitive Fatigue Assessment Platform

Jul 2026 – Present

- Designing and building a TypeScript/React, FastAPI, and PostgreSQL platform that measures cognitive fatigue through three validated behavioral tasks (Psychomotor Vigilance, 2-back, Stroop), collected 2–3x daily to build a longitudinal, trial-level dataset for personalized fatigue modeling.
- Currently collecting real-world data ahead of an XGBoost baseline and transformer-based model.

ML Algorithms from Scratch

Apr 2026 – Present

- Implemented a Transformer (encoder + decoder, per "Attention Is All You Need"), a 3-layer Multilayer Perceptron, and an XGBoost model from mathematical first principles with no ML libraries. All three articles were published in Towards AI (150K followers) on Medium.

Computer Vision Tracking System

Apr 2025 – Jun 2025

- Applied YOLOv8 transfer learning in Python on 4,000+ manually annotated objects (players, referees, goalies) to build a player-tracking system from raw game footage, achieving 0.917 mAP50 (0.569 mAP50-95).

RESEARCH

Democratizing ESG (team of 3)

Feb 2026 – May 2026

- Built an NLP pipeline in Python to predict ESG scores for 323 S&P companies from free public data only, combining FinBERT sentiment, keyword, and readability features from SEC 10-K filings with EDGAR financials; ElasticNet outperformed XGBoost (CV R^2 0.19 vs. 0.08) and reduced 130 features to ~20 interpretable predictors.
- Diagnosed a bimodal disclosure threshold in the ratings as the main performance ceiling; an oracle experiment showed correct tier assignment would raise environmental R^2 from 0.02 to 0.74.

Undergraduate Thesis: Injury Impact on Athlete Careers

Sep 2024 – Apr 2025

- Built a dataset of 472 player-seasons across 5 ACC women's lacrosse programs (2015–2024) by scraping team sites and manually coding injuries from player bios; clustered players into 7 roles with K-Means and trained multi-output XGBoost in Python with lag features (R^2 0.76, RMSE 0.56 for next-season goals per game) to simulate injury-free counterfactual careers.

Market Efficiency Analysis (team of 3)

Sep 2023 – Feb 2024

- Tested sportsbook efficiency across 10,000+ games (2016–2022) using regression-based efficiency tests and backtests of 8 wagering strategies plus win-streak ("hot hand") analysis; the best strategy returned under 0.1% ROI on \$890K in simulated wagers. Presented at the Academy of Economics and Finance Conference (Feb 2024).

SKILLS

Programming Languages: Python, R, SQL

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Transformers, NLP, YOLOv8

Cloud & Data: Azure, AWS, PostgreSQL, Snowflake

Other: Git/GitHub, Excel, Tableau, Power BI, Pandas, NumPy, Claude Code